

Student Academic Performance Analysis

Data Analytics Project using Python, Pandas, Matplotlib and Seaborn

Project Objective: To analyze student academic data and identify patterns and factors associated with examination performance.

Dataset records	6,607 students
Number of columns	20
Main outcome	Exam_Score
Programming language	Python
Environment	Google Colab
Main libraries	Pandas, NumPy, Matplotlib, Seaborn

1. Introduction

Student academic performance can be influenced by several academic, personal, family and school-related factors. This project uses student-level data to explore how these factors are associated with exam scores. The analysis focuses on data cleaning, descriptive statistics, correlation analysis, group-wise comparisons and visualization.

2. Problem Statement

Educational institutions collect information about attendance, study habits, previous performance, resources, motivation, parental involvement and other student characteristics. The purpose of this project is to analyze these variables and identify useful patterns related to examination performance.

3. Objectives

- Understand the structure and quality of the student dataset.
- Identify and handle missing values.
- Check for duplicate records and basic data-quality issues.
- Study relationships between numerical factors and exam scores.
- Compare average exam scores across important categorical factors.
- Create clear visualizations and summarize actionable observations.

4. Dataset Description

The dataset contains 6,607 student records and 20 variables. Each row represents one student and each column represents an academic, personal, family or school-related attribute. Exam_Score is used as the main outcome variable.

Column	Meaning
Hours_Studied	Number of hours spent studying
Attendance	Student attendance percentage
Previous_Scores	Previous academic score
Tutoring_Sessions	Number of tutoring sessions
Exam_Score	Exam performance / target outcome
Parental_Involvement	Level of parental involvement
Access_to_Resources	Level of access to educational resources
Motivation_Level	Student motivation level
Teacher_Quality	Recorded teacher quality level
Peer_Influence	Influence of peers on the student
Learning_Disabilities	Recorded learning-disability indicator
Distance_from_Home	Distance category from home to school

5. Data Cleaning and Preparation

Three categorical columns contained missing values: Teacher_Quality (78), Parental_Education_Level (90), and Distance_from_Home (67). These missing values were represented as 'Missing' rather than deleting the corresponding student records. After cleaning, no missing values remained. A duplicate check found 0 duplicate records.

One source record contains an Exam_Score of 101. It was retained rather than arbitrarily changed, and it should be treated as a data-quality observation when interpreting the results.

6. Exploratory Data Analysis

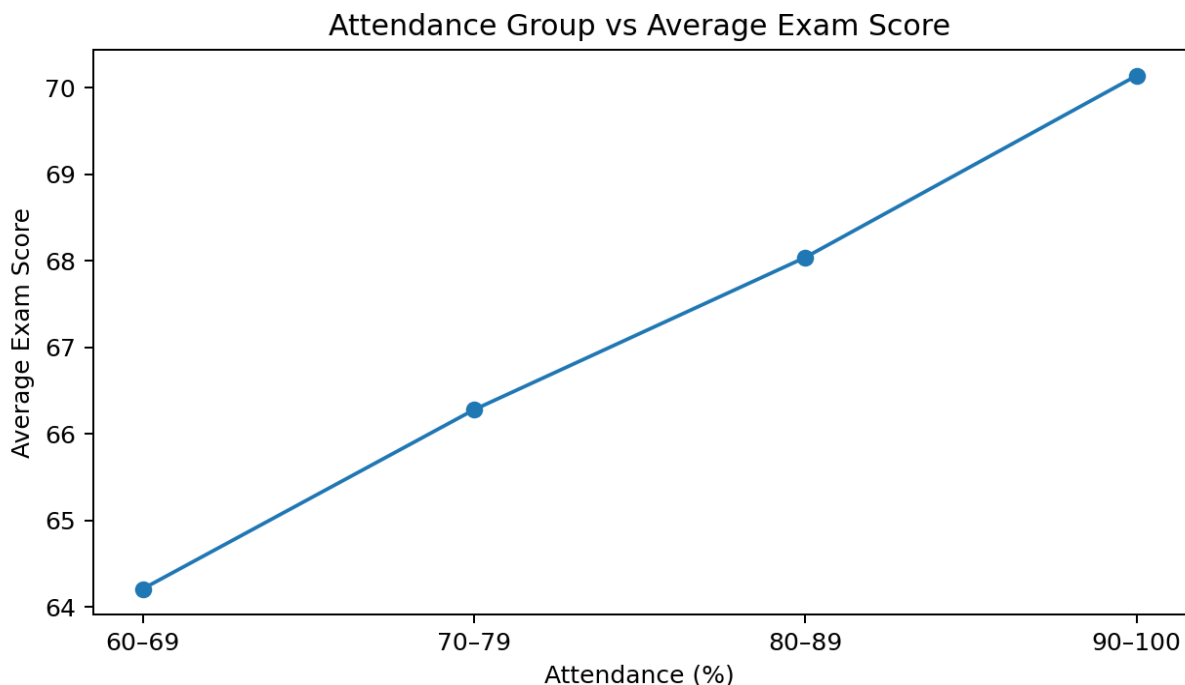
Descriptive statistics showed an average Exam_Score of approximately 67.24. Average attendance was about 79.98%, while average study time was about 19.98 hours in the dataset. Correlation analysis was used to quantify linear association between numerical variables and Exam_Score.

Numerical factor	Correlation with Exam_Score
Attendance	0.581
Hours_Studied	0.445
Previous_Scores	0.175
Tutoring_Sessions	0.157
Physical_Activity	0.028
Sleep_Hours	-0.017

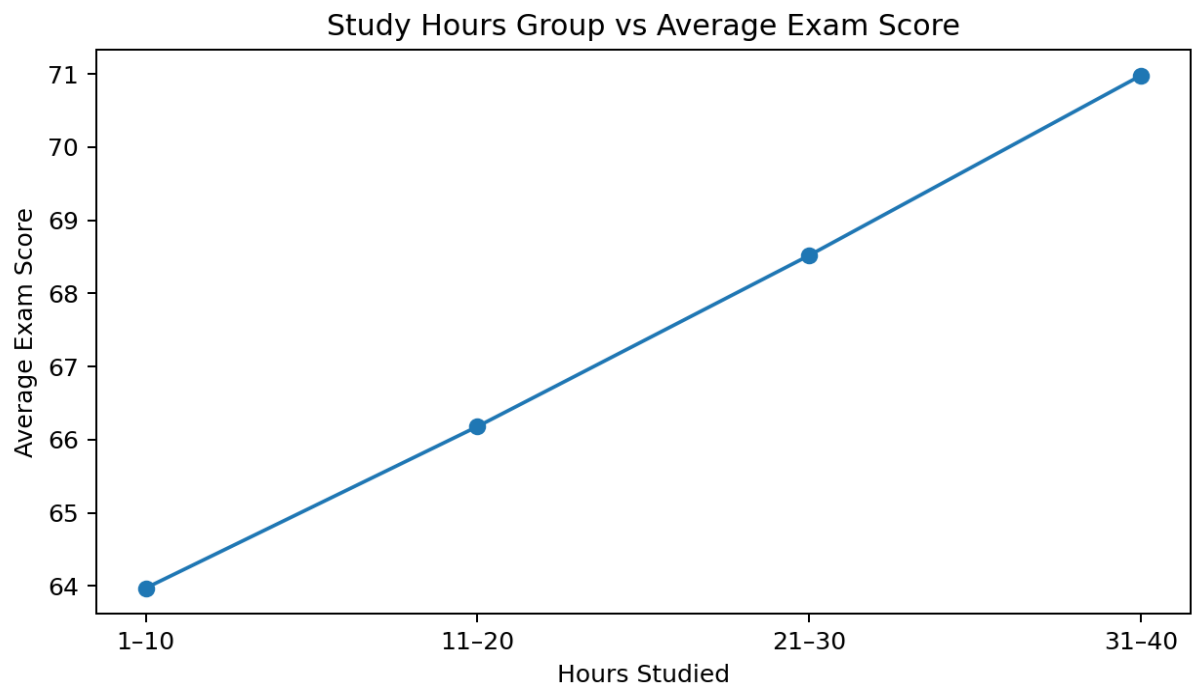
Interpretation: Attendance showed the strongest positive linear association with Exam_Score among the numerical variables, followed by Hours_Studied. These are associations and should not be interpreted as proof of causation.

7. Visualizations and Findings

Attendance: Students in the 90–100% attendance group had an average score of 70.14, compared with 64.21 for the 60–69% group.

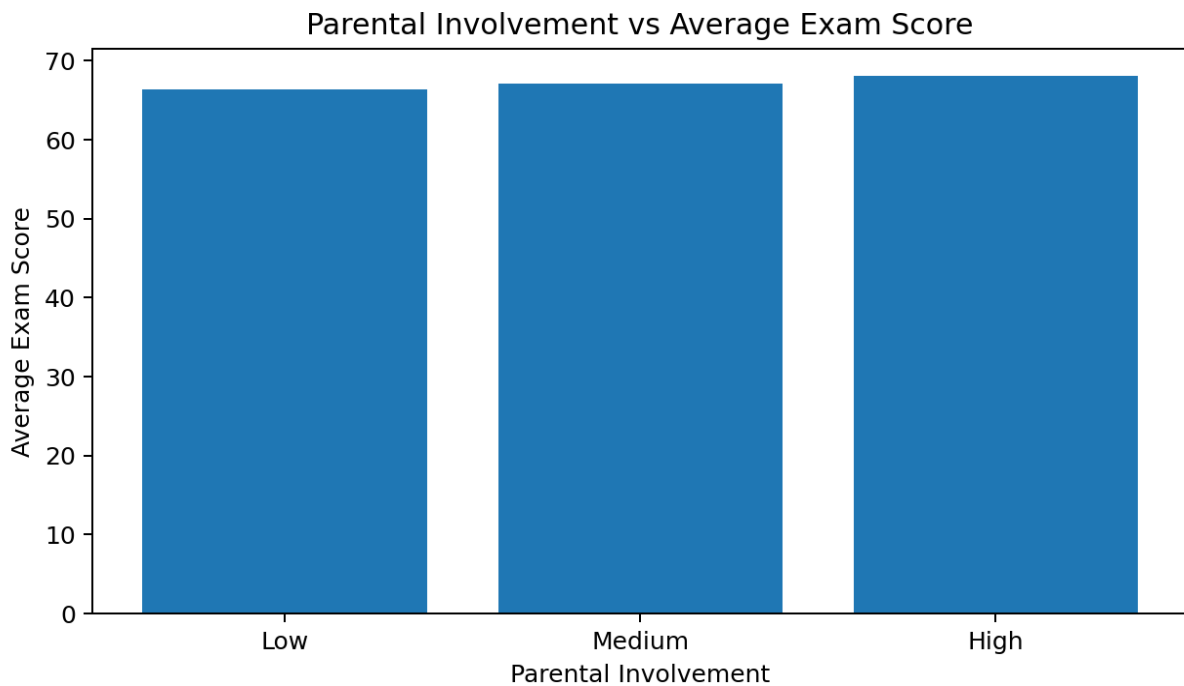


Hours Studied: Average scores generally increased across the study-hour groups. The 31–40 hour group had an average score of 70.98. The 41–50 group contains only two records, so it should not be generalized.

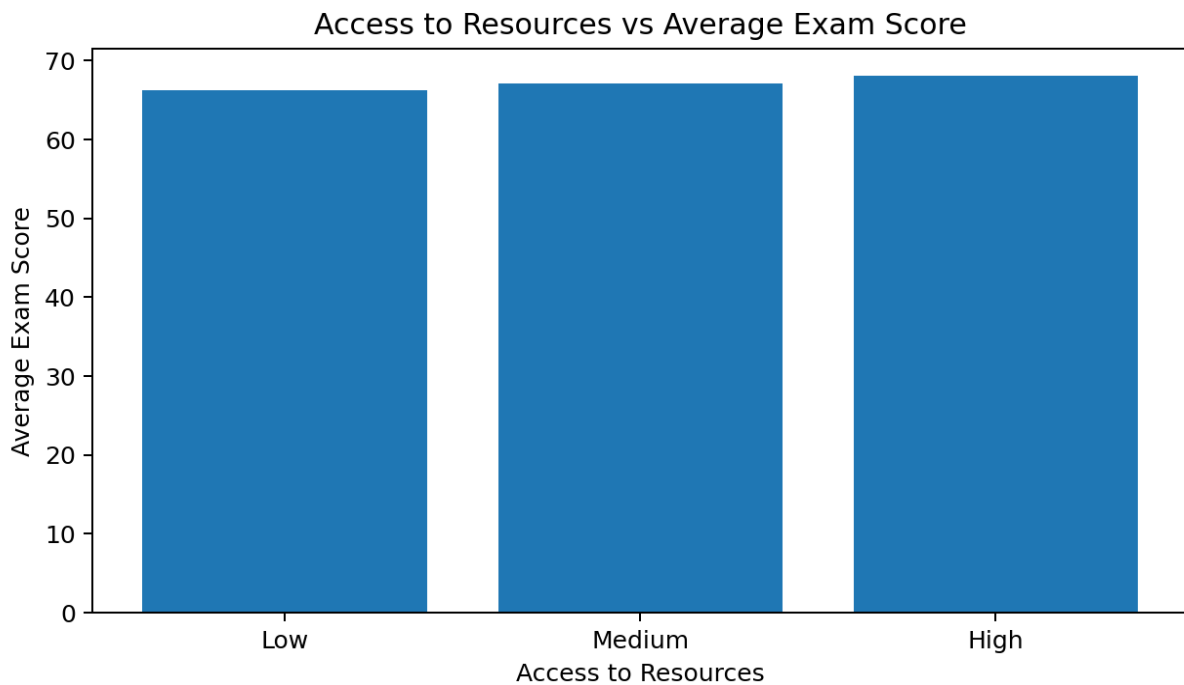


8. Categorical Factor Analysis

Parental Involvement: Average Exam_Score was 66.36 for Low, 67.10 for Medium and 68.09 for High parental involvement.



Access to Resources: Average Exam_Score was 66.20 for Low, 67.13 for Medium and 68.09 for High access to resources.



9. Other Observations

- Higher motivation levels were associated with slightly higher average exam scores.
- Higher recorded teacher quality was associated with slightly higher average scores.

- Positive peer influence was associated with a higher average score than neutral or negative peer influence.
- Students with internet access had a slightly higher average score than students without internet access.
- The analysis includes both numerical and categorical variables to provide a broader view of student performance.

10. Real-World Application

Schools and educational institutions can use similar analytics to identify patterns in student performance and areas where additional academic support or resources may be useful. The analysis can support data-informed discussions, while decisions about individual students should consider context beyond the variables in this dataset.

11. Tools and Technologies

Python was used as the main programming language. Pandas was used for data loading, cleaning and analysis; NumPy for numerical operations; Matplotlib and Seaborn for visualization; and Google Colab as the development environment.

12. Conclusion

The project analyzed 6,607 student records to understand factors associated with academic performance. Attendance and Hours_Studied showed the strongest positive numerical associations with Exam_Score in this dataset. Group comparisons also showed differences across parental involvement, access to resources, motivation, teacher quality and peer influence. The project demonstrates a complete basic data analytics workflow from data cleaning and exploration to visualization and interpretation.

13. Future Scope

- Build a prediction model for exam performance.
- Develop an interactive Power BI or Tableau dashboard.
- Perform statistical significance testing for selected group differences.
- Collect additional real-world student data for validation.
- Explore more advanced machine-learning and explainability techniques.

Note: Correlation and group-wise averages describe patterns in the available data; they do not establish that a particular factor causes higher or lower exam scores.